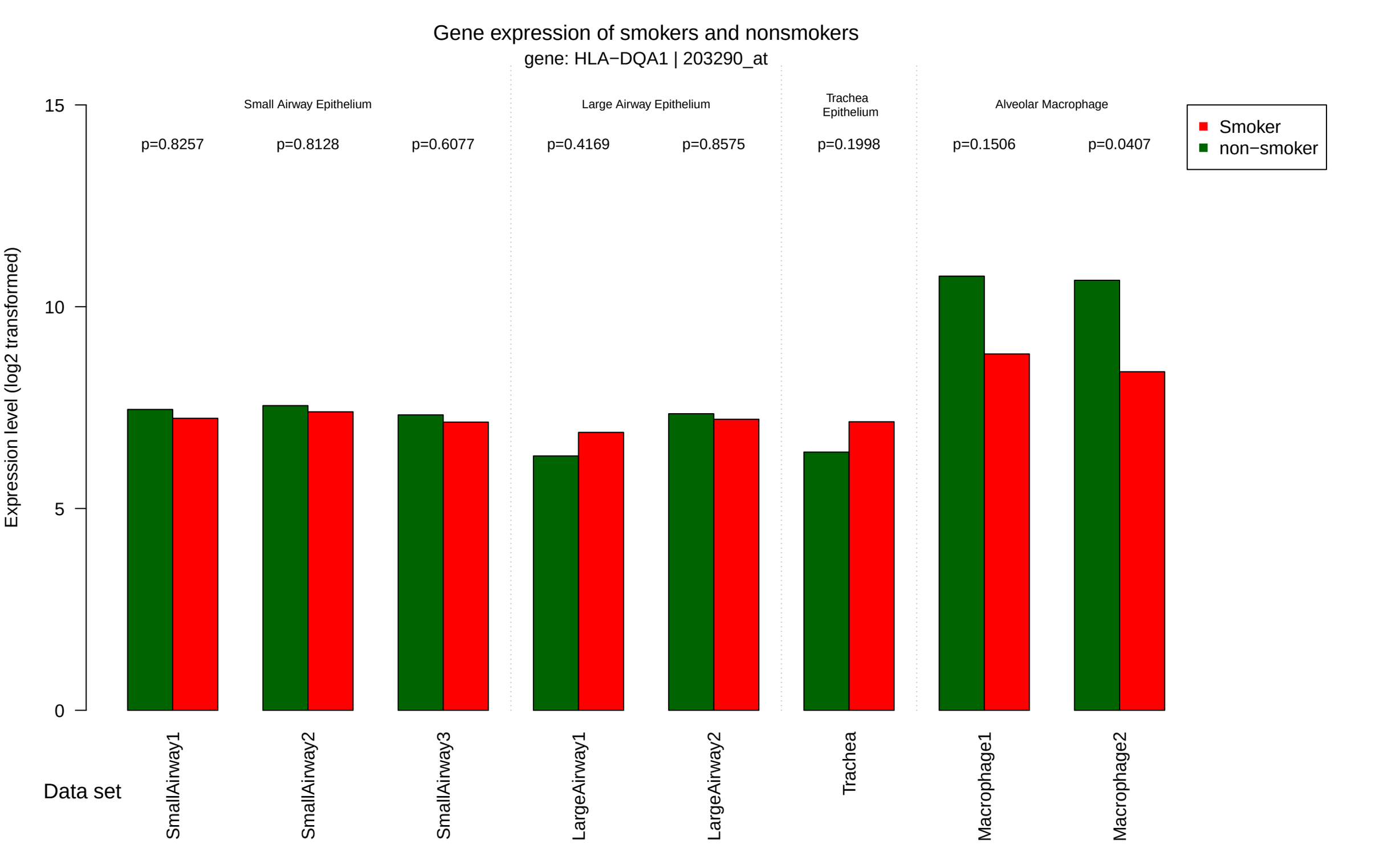
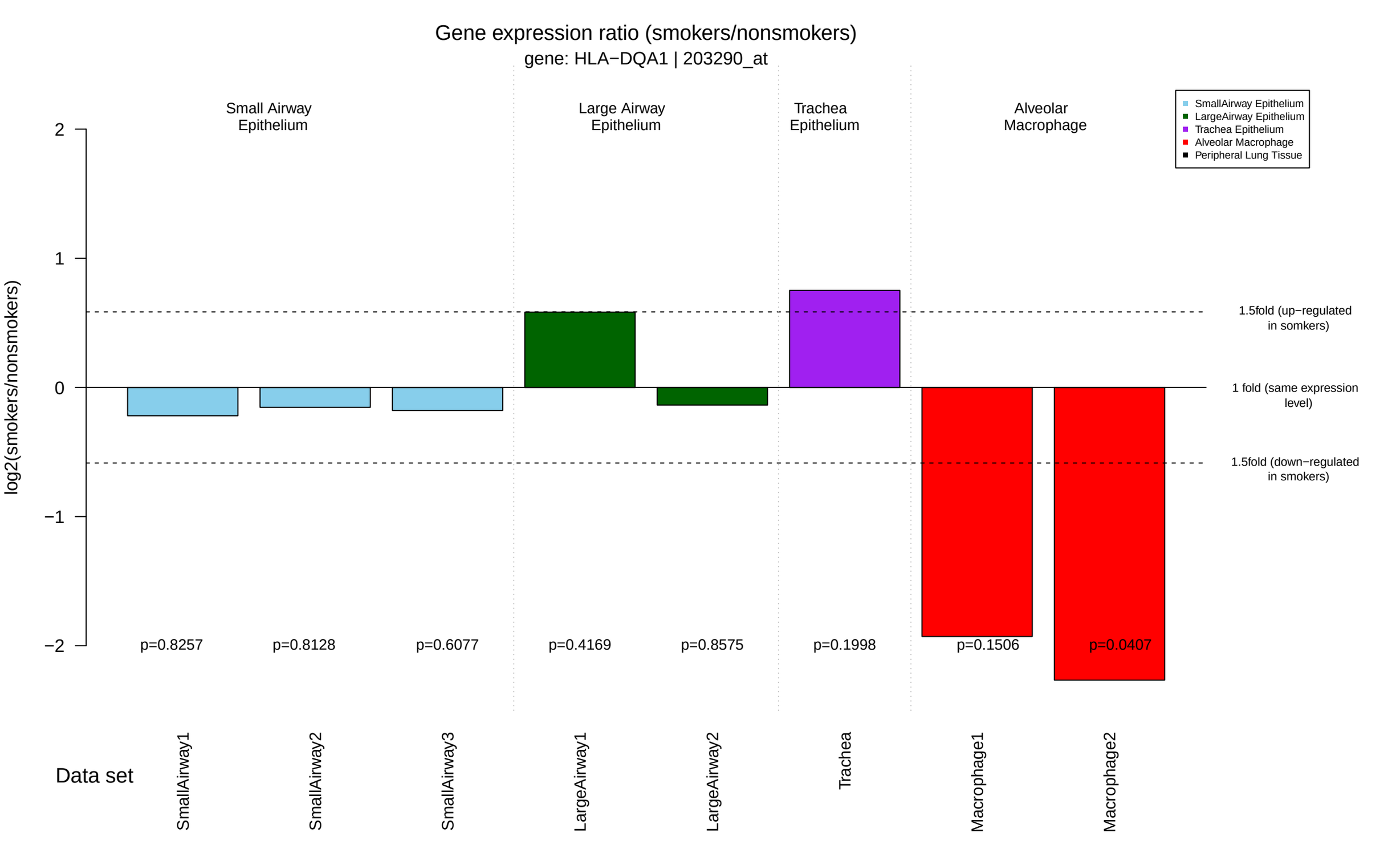
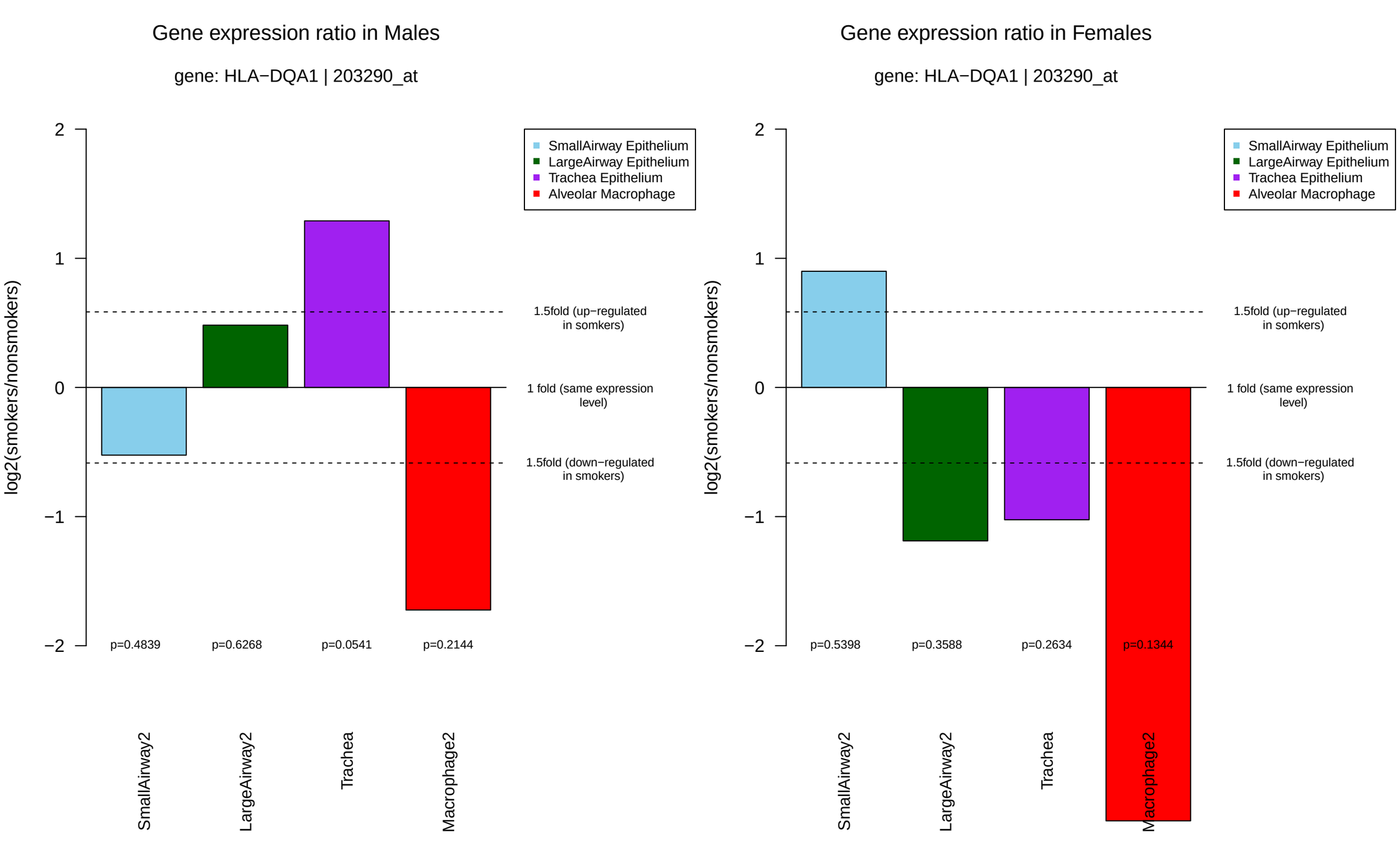
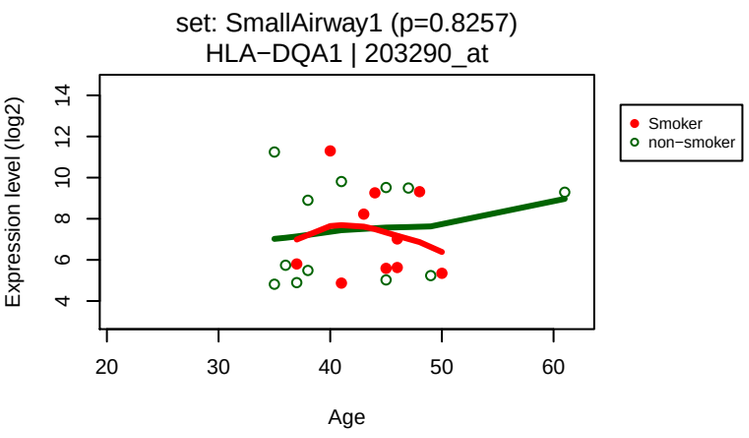
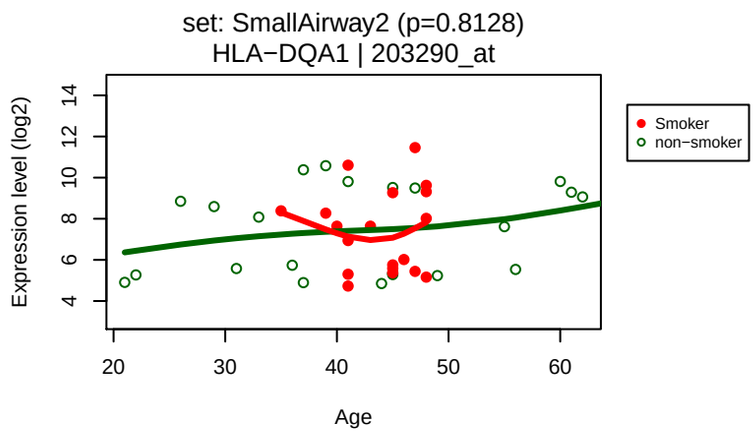
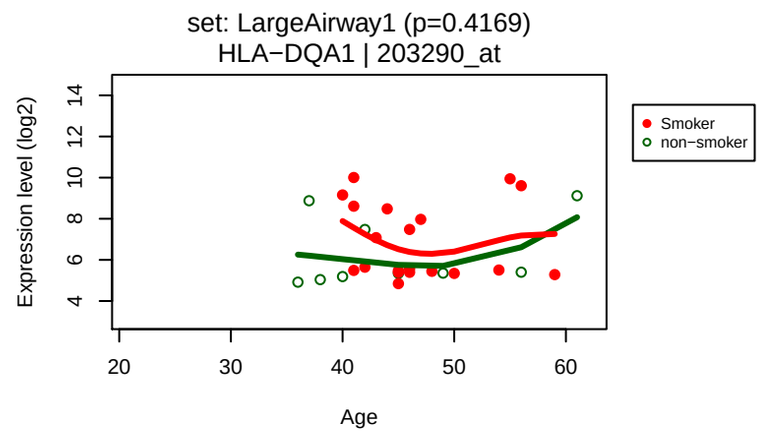

Gene expression of smokers and nonsmokers
gene: HLA−DQA1 | 203290_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8257
p=0.8128
p=0.6077
p=0.4169
p=0.8575
p=0.1998
p=0.1506
p=0.0407
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

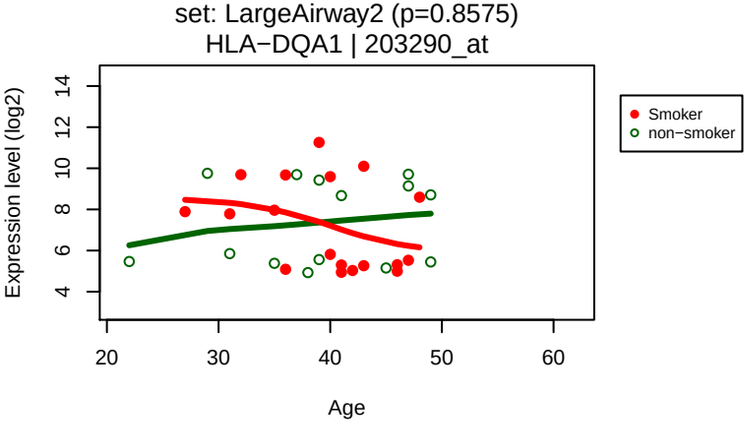

Gene expression ratio (smokers/nonsmokers)
gene: HLA−DQA1 | 203290_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8257
p=0.8128
p=0.6077
p=0.4169
p=0.8575
p=0.1998
p=0.1506
p=0.0407
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

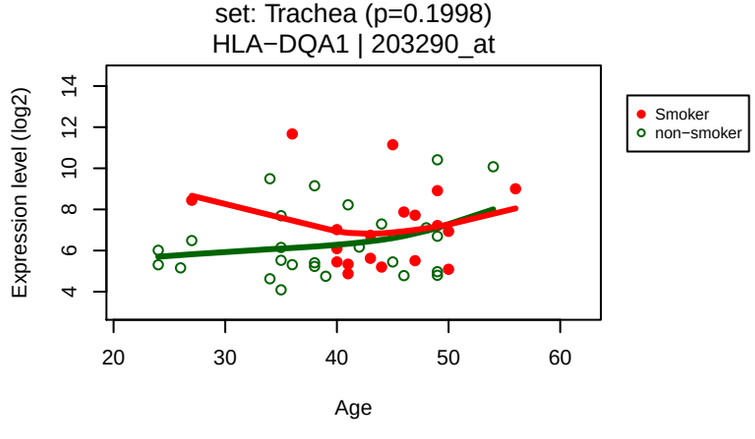

Gene expression ratio in Males
gene: HLA−DQA1 | 203290_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4839
p=0.6268
p=0.0541
p=0.2144
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HLA−DQA1 | 203290_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5398
p=0.3588
p=0.2634
p=0.1344
SmallAirway2
LargeAirway2
Trachea
Macrophage2

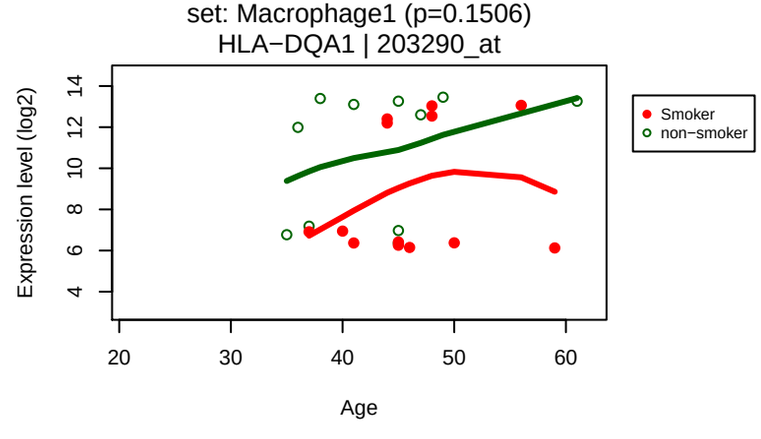

set: SmallAirway1 (p=0.8257)
HLA−DQA1 | 203290_at
Expression level (log2)
Age
Smoker
non−smoker

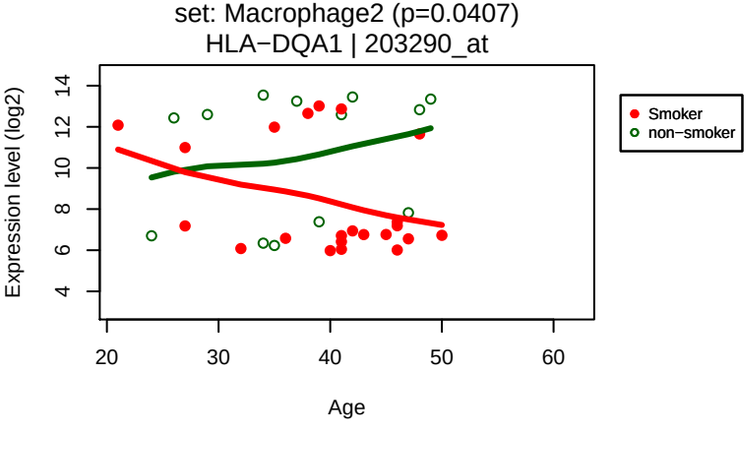

set: SmallAirway2 (p=0.8128)
HLA−DQA1 | 203290_at
Expression level (log2)
Age
Smoker
non−smoker

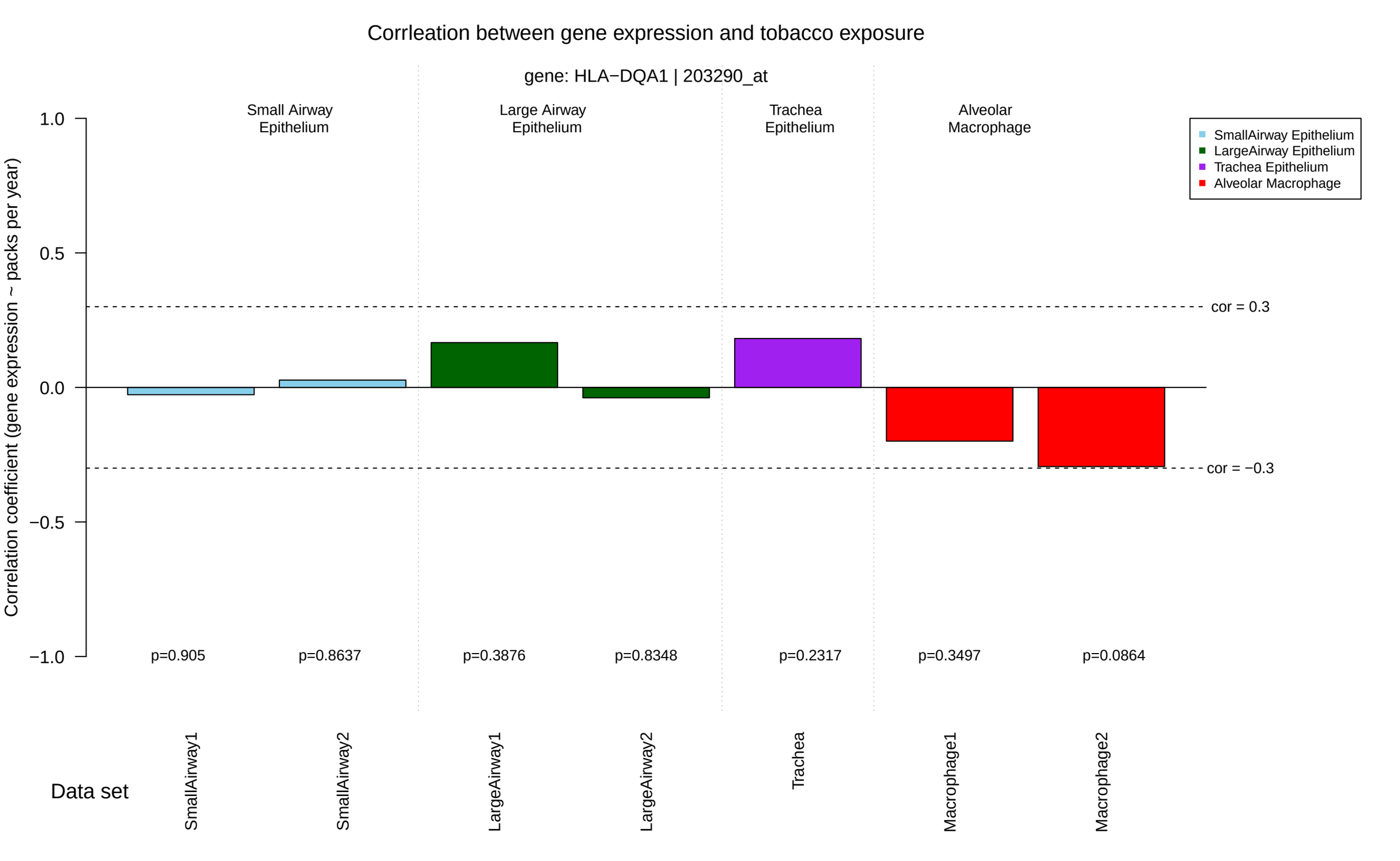

set: LargeAirway1 (p=0.4169)
HLA−DQA1 | 203290_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8575)
HLA−DQA1 | 203290_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1998)
HLA−DQA1 | 203290_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1506)
HLA−DQA1 | 203290_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0407)
HLA−DQA1 | 203290_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HLA−DQA1 | 203290_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.905
p=0.8637
p=0.3876
p=0.8348
p=0.2317
p=0.3497
p=0.0864
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2